

Permeability anisotropy induced by the shear displacement of rough fracture walls

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[1] The permeability anisotropy that results from a shear displacement $\vec{u}$ between the complementary self-affine walls of a rough fracture is investigated. Experiments in which a dyed fluid radially injected into a transparent fracture displaces a transparent one are presented. A clear anisotropy is observed in the presence of shear displacements and allows us to estimate the ratio of the permeabilities for flows parallel and perpendicular to $\vec{u}$. A simple model which accounts for the development of channels perpendicular to $\vec{u}$ qualitatively explains these results and predicts a permeability decreasing (increasing) linearly with the variance of the aperture field for flow parallel (perpendicular) to the shear displacement. These predictions are then compared to the results of numerical simulations performed using a lattice Boltzmann technique and to the anisotropies measured in displacement experiments.

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1. Introduction

[2] Subsurface fluid flow in many low-permeability geological formations occurs primarily through fracture networks [NAS Committee on Fracture Characterization and Fluid Flow, 1996; Adler and Thovert, 1999; Sahimi, 1995]. In order to model such systems, one needs to understand flow in single fractures, which are the building blocks of the network. Among the many parameters which influence fluid motion at this scale, we focus here on the geometry of the fractures, specifically on the effect of correlations between the roughness of the two opposite walls on the permeability of the fracture. The two surfaces completely match each other, but a shear displacement of one of the surfaces alters these correlations, and previous laboratory measurements demonstrated that it also affects the permeability, through the corresponding variations of the local aperture of the fracture [Plouraboué et al., 2000; Olsson and Brown, 1993].

[3] In related experiments, a higher permeability was measured for flow perpendicular to the shear displacement than parallel to it [Gentier et al., 1997; Yeo et al., 1998], and an indication of such anisotropic permeability was also observed in previous work on self-affine fractures [Drazer and Koplik, 2002]. Numerical computations performed in fractures with anisotropic aperture fields showed a similar behavior in the permeability [Thompson and Brown, 1991]. The purpose of the present paper is to combine experiments, numerical simulations, and analytical arguments to understand the permeability anisotropies that are induced by shear displacements of rough fracture walls.

[4] A key geometrical feature of the fracture surfaces studied in this work is their statistical scale invariance known as self-affinity [Mandelbrot, 1983; Feder, 1988]. The analysis of the surfaces of both faults and fresh fractures has established that their roughness cannot be described by a finite set of typical wavelength values, but instead is self-affine. A self-affine surface is one that is statistically unchanged under the scaling relation: $x \rightarrow \lambda x$, $y \rightarrow \lambda y$ and $h(x, y) \rightarrow \lambda^\zeta h(x, y)$. ((x, y) are the coordinates in the mean surface plane, $h(x, y)$ is the local height of the surface, and ζ is known as the Hurst exponent.) The self-affine character of the fracture surfaces has been observed, over a significant range of length scales, in a broad variety of materials and for both natural and man-made fractures [Bouchaud, 2003]. In the case of fractured rocks, such as granite or basalt rocks, the Hurst exponent is found to be $\zeta \approx 0.8 \pm 0.05$ over a broad range of length scales [Bouchaud, 2003], although values near 0.5 are found in other rocks [Boffa et al., 1998].

[5] The key consequence of self-affinity in our context is the following scaling law for the variance of the surface heights:

$$\left\langle \left[h(\vec{x} + \vec{\lambda}) - h(\vec{x}) \right]^2 \right\rangle \sim |\vec{\lambda}|^{2\zeta}, \quad (1)$$

which implies long-range correlations in the height-to-height fluctuations. We shall show that these correlations strongly affect the aperture field of the rough fractures studied in the present work and therefore flow inside them.

[6] In both experiments and numerical studies, we shall consider model fractures, made of two complementary self-affine surfaces: two perfectly matching surfaces separated by a distance a_0 in the direction normal to their common mean plane and, in general, subject to a shear displacement $\vec{u}$ parallel to this mean plane. As pointed out in previous studies [Brown et al., 1986; Plouraboué et al., 1995], the shear displacement introduces a mismatch length between the two perfectly mated surfaces and the self-affine scaling behavior seen in the topography of an individual surface

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will only appear in the aperture field at length scales smaller than the shear displacement. In all cases, the gap between surfaces is large enough so that the two walls do not touch, thereby avoiding the effects of shear-induced dilation of the fracture gap on its permeability.

[7] The experiments were performed using transparent models fabricated from moldings of actual rocks. In the experiments, a dye is injected radially from a point of the fracture and the anisotropy in the permeability is computed from that of the injection patterns. Only the ratio of the highest and lowest permeabilities can be determined in this way, but not their absolute values. These experiments are therefore appropriately complemented by numerical simulations which, in addition to the permeability, provide the complete flow field. In order to compute the velocity field in these complex geometries we used the lattice Boltzmann technique [Dräger and Koplik, 2000, 2001, 2002; Kim *et al.*, 2003].

[8] The paper is organized as follows: in section 2.1 we describe the experimental procedure used to obtain the transparent cast of the fractured rock. Also in section 2.1 we describe the method used to characterize the roughness of the fracture walls. In section 2.2 we describe the experimental setup and the image analysis method. In section 3 we present the experimental results and, in section 4, we discuss a theoretical analysis of the relation between the aperture field and the observed anisotropy in the permeability. The scaling laws obtained in this way are then compared to numerical simulations in sections 5 and 6. Finally, the theoretical description developed in section 4 is applied to the experimental data in section 7.

2. Experimental Setup and Procedure

2.1. Model Fracture and Flow Setup

[9] The experiments were performed on transparent moldings of fractured granite blocks extracted from a quarry in Lanhelin (Britanny). A tensile crack was produced by a compressive force applied on two opposite sharp edges carved in the middle of two facing sides (the initial block size is $25 \times 25 \times 40 \text{ cm}^3$). A silicon rubber molding of one of the fracture surfaces is then used as a template to produce a transparent epoxy casts of the original surface. The epoxy cast is in turn used as a template to produce a second silicon rubber molding of a perfectly matching surface (dimensional deformations during polymerization are much smaller for the silicon than for the epoxy). This method is in common use, and typical reproduction problems are negligible in the present type of application [Hakami and Barton, 1990; Yeo *et al.*, 1998; Auradou *et al.*, 2001; Isakov *et al.*, 2001; Kostakis *et al.*, 2003]. The complementary epoxy and rubber surfaces are the walls of the model fracture used in the present work.

[10] Prior to the experiments, 2-D maps of the epoxy casts of different fractures are obtained by means of a mechanical profilometer (770×760 points), with a $250 \text{ }\mu\text{m}$ grid spacing and a $10 \text{ }\mu\text{m}$ vertical resolution [Boffa *et al.*, 1998]. The difference in height between the highest and the lowest points of the surface is $\simeq 17 \text{ mm}$ and the root mean square deviation of the surface heights is $\simeq 3.5 \text{ mm}$. As discussed in the introduction, self-affine surfaces have a roughness which does not contain any intrinsic wavelength, and it

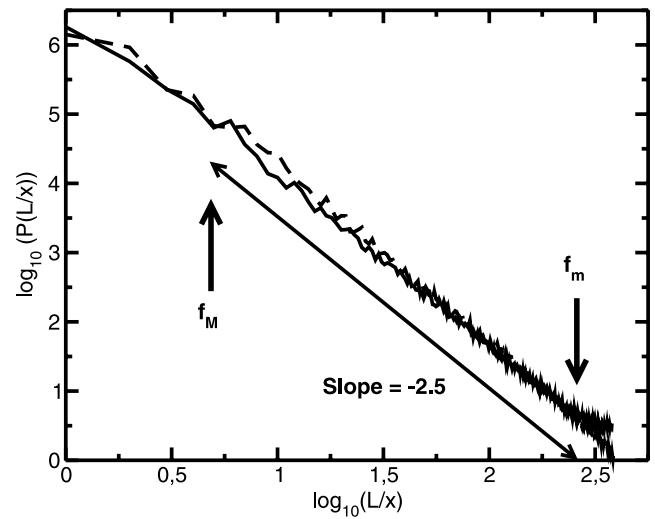


Figure 1. Power spectral density of the experimental fracture surfaces. The spatial frequencies are $f = L/(n\Delta x)$, for $n = 1, 2, \dots, L/\Delta x$, where $L = 20 \text{ cm}$ is the fracture length and $\Delta x = 250 \text{ }\mu\text{m}$ is the resolution length along the profile. The solid and dashed lines correspond to profiles obtained normal and parallel to the direction of crack propagation. The straight line has a slope -2.5 .

can be shown that the power spectral density $P(f)$ of the heights, for a linear profile drawn on the surface, obeys the following scaling law with the spatial frequency f [Feder, 1988]:

$$P(f) \sim Af^{-1-2\zeta}. \quad (2)$$

Such spectral densities $P(f)$ obtained for our fracture surfaces are plotted in Figure 1 and follow, indeed, a power law as a function of f . The computed spectral densities showed no dependence, within experimental error, on the orientation of the profile used to compute them: this implies that the fractures are isotropic in their mean plane. Note that this situation differs from previous studies in which the fracture surface itself was anisotropic [Thompson and Brown, 1991]. Here, the anisotropy in the aperture field only appears as a result of the shear displacement between the opposite surfaces of the fracture. By averaging spectra obtained for several orientations of the profiles, one obtains $\zeta = 0.75 \pm 0.05$, in agreement with previous studies of granite fractures [Boffa *et al.*, 1998; Méheust and Schmittbuhl, 2000; Auradou *et al.*, 2001]. As indicated by the solid line with arrows at both ends in Figure 1, $P(f)$ follows this power law over approximately two decades: the corresponding length scales range from 50 mm down to 0.75 mm . The lower limit f_m of this range is set by the resolution of the present map; previous measurements have shown that the self-affine description remains valid at lower length scales, of the order of 0.1 mm or less [Boffa *et al.*, 1998]. The upper limit f_M arises from the limited number of wavelengths which describe the surface topography at large scales: f_M is thus related to the finite size of the sample, which has a large-scale topography dominated by a few large wavelengths. This statistical

effect usually becomes apparent at length scales greater than one fourth of the system size [Méheust and Schmittbuhl, 2000].

2.2. Experimental Procedure and Image Analysis

[11] The objective of the experiments is to analyze flow inside an open fracture by monitoring the radial spreading of an injected dye. Using a radial injection geometry allows us to estimate variations of the fracture permeability with the flow direction from the shape of the invaded zone. The two complementary surfaces are initially brought into full contact, with their mean plane (x, y) horizontal. The top (transparent) surface is kept fixed thereafter (see Figure 2 for details) and the lower one can be translated while remaining parallel to its initial orientation. Specifically, it is first displaced downward ($-z$ direction), by a distance a_0 , to open the fracture, and then laterally in the (x, y) plane to the desired location. This lateral displacement $\vec{u}$ is small enough so that the two surfaces do not come in contact. In order to ensure an isotropic boundary condition at the outlet, the rubber molding has been cut to a circular shape with its center at the injection point.

[12] The fracture model is then placed inside a leak tight basin which is filled with transparent fluid in order to saturate the model prior to the experiments. Transparent fluid is then injected through a hole in the top surface and flows radially outward. Once a steady flow is reached, a dyed fluid is injected through the inlet orifice, and images of the invaded region are taken at constant time intervals using a high resolution cooled 12 bits digital camera. The cell is uniformly illuminated from above by four halogen lamps placed around the setup and the light reflected by the opaque silicon cast, which corresponds to the bottom surface of the fracture (see Figure 2), is recorded by the camera. Each pixel corresponds to a $250 \mu\text{m} \times 250 \mu\text{m}$ area. The fluids used are water-glycerol solutions with 10% glycerol and viscosity 1.288 cP at 20°C [Lide, 1996]. The dyed solution contains 0.2–0.3 g/L of nigrosine and the density of the two fluids is matched by adding an equivalent amount of NaCl to the transparent solution. In order to avoid density differences due to temperature variations the solutions are stored at the same temperature. In all experiments, density differences between the two fluids were always smaller than $25 \cdot 10^{-4} \text{ g/cm}^3$ [Lide, 1996] and the flow rate is less than 5 mL/min. The Reynolds number is less than 1 even in the vicinity of the injection.

[13] The concentration map of the injected dye is then determined from the images recorded during the experiments, by assuming the validity of Beer-Lambert's law [Brown et al., 1998; Detwiler et al., 1999, 2000], i.e., that the light intensity $I(x, y)$ at any given point (x, y) of the image varies exponentially with the local aperture $a(x, y)$ and the local concentration $c(x, y)$ according to

$$I(x, y) = I_0(x, y) \exp(-2 \mu c(x, y) a(x, y)), \quad (3)$$

where $I_0(x, y)$ is the light intensity for a transparent fluid and μ is the extinction coefficient. The validity of this relation has been verified by opening the fracture without introducing a lateral shift so that the local aperture $a(x, y)$ is constant and equal to the vertical displacement Δz . The corresponding light intensity $I(x, y)$ was then determined for different dye concentrations, c_0 , and different mean

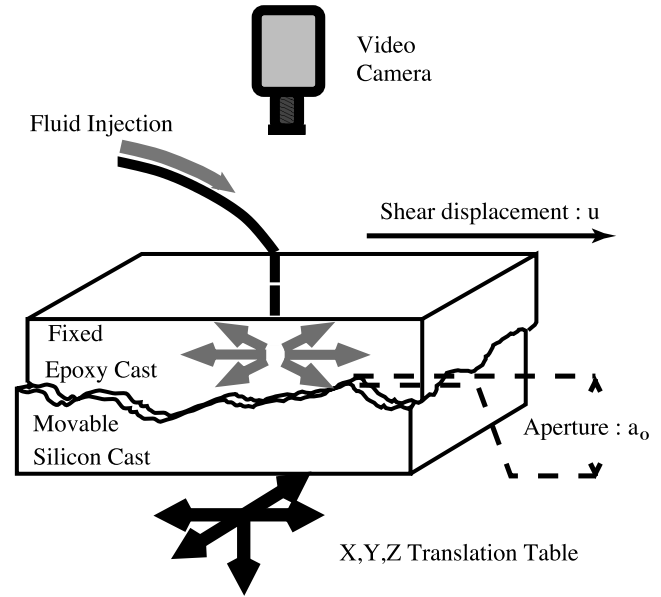


Figure 2. Schematic view of the experimental setup. The top wall is a transparent epoxy cast and has a fixed position. The bottom wall is a silicone replica of the top wall and can be moved down and laterally. The dyed fluid is injected in a radial, divergent configuration, and it displaces a nondyed fluid which initially saturates the fracture gap. The flow process is visualized with a camera set above the fracture. The two controlled lengths are indicated: a_0 , the mean vertical separation between walls, and $\vec{u}$, the shear displacement vector.

apertures, $a_0 = \Delta z$. The corresponding results approximately collapse into a single curve by plotting the ratio $I(x, y)/I_0(x, y)$ as a function of $\Delta z \times c_0$. Moreover, the variation of the absorbance with $\Delta z \times c_0$ can be approximated, in a log linear plot, by the linear relationship that follows from equation (3) with $\mu = 2.9 \pm 0.4 \text{ m}^2/\text{g}$. The previous linear dependence of the absorbance is valid for $a(x, y) c_0 < 0.15 \text{ g/m}^2$. The uncertainty on the extinction coefficient μ mostly reflects the departure from this linear law and has been previously observed with this experimental technique [Detwiler et al., 1999, 2000]. Then, we can determine the local, instantaneous concentration $c(x, y, t)$ during the displacement experiments, by also applying equation (3) to the asymptotic concentration inside the fracture:

$$\frac{c(x, y, t)}{c_f} = \ln \left[\frac{I(x, y, t)}{I_0(x, y)} \right] / \ln \left[\frac{I_0(x, y)}{I_f(x, y)} \right], \quad (4)$$

where c_f is the asymptotic concentration and I_f is the corresponding final light intensity. A typical concentration map obtained in this way for an experiment in which the fracture walls are translated relative to each other is displayed in Figure 3a. The maximum RMS error was 6% for any location inside the fracture. In addition to the irregularity of the front boundary, already discussed by Auradou et al. [2001] and by Drazer et al. [2004], its global shape is clearly anisotropic, with a strong elongation in the direction perpendicular to the shear displacement $\vec{u}$. A similar anisotropy is observed on the aperture field

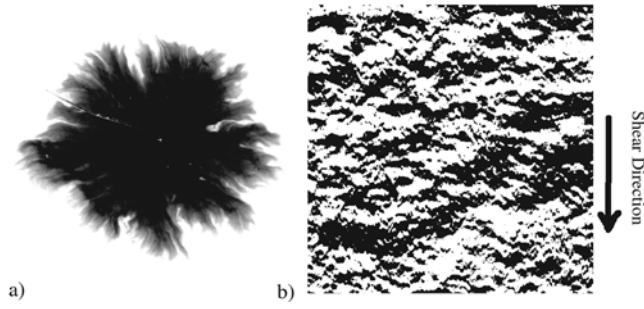


Figure 3. Example of the relation between flow anisotropy and aperture field in a sheared fracture, with $a_0 = 1.25$ mm and $u_y = 2.0$ mm (the direction of the shear displacement is indicated by the arrow). The normalized aperture variance is $S^2 = 0.133$. (a) Concentration map. (b) Aperture field computed using equation (4), where white corresponds to a local aperture $a(x, y) < a_0$ and black corresponds to $a(x, y) > a_0$.

displayed in Figure 3b, with the appearance of elongated channels, which are also perpendicular to $\vec{u}$.

2.3. Quantitative Characterization of Anisotropy and Relation to Permeability Field

[14] The anisotropy of the invaded zone can be characterized quantitatively from the inertia tensor $\mathbf{M}(t)$ of the concentration field:

$$\mathbf{M}(t) = \frac{1}{c_f} \sum c(x, y, t) \begin{pmatrix} y_*^2 & -x_* \times y_* \\ -x_* y_* & x_*^2 \end{pmatrix}, \quad (5)$$

The sum is taken over all pixels in the image and $x_* = x - x_c$ and $y_* = y - y_c$, where (x_c, y_c) is the center of mass of the invasion pattern. The eigenvalues λ_+^2 and λ_-^2 of the inertia tensor $\mathbf{M}(t)$ and the corresponding principal axes are then determined: they characterize the magnitude of the anisotropy and the directions of fastest and slowest flow, respectively.

[15] More quantitatively, the tensor $\mathbf{M}(t)$ can be used to estimate the ratio between the permeabilities for flow along these two directions. Let us ignore the surface roughness and consider a two dimensional Hele-Shaw flow with a weak anisotropy of the permeability so that $k_x \neq k_y$, but $|k_x - k_y| \ll (k_x + k_y)$; it is then possible to compute the average elliptical shape of a convected tracer front as follows. Let us assume that a dyed fluid is injected through a small hole of radius r_0 at pressure p_0 , and allowed to escape at the circular outer boundary $r = r_1$ at atmospheric pressure $p_1 \equiv 0$. Then, the local velocities are given by

$$v_x = k_x - \frac{\partial p}{\partial x} \quad \text{and} \quad v_y = k_y - \frac{\partial p}{\partial y}, \quad (6)$$

which together with the incompressibility condition $\nabla \cdot \vec{v} = 0$ gives

$$k_x \frac{\partial^2 p}{\partial x^2} + k_y \frac{\partial^2 p}{\partial y^2} = 0. \quad (7)$$

The pressure is readily found to be, to leading order in the permeability contrast,

$$p(r, \theta) = \frac{p_0}{\ln(r_0/r_1)} \left[\ln(r/r_1) + \frac{k_x - k_y}{2(k_x + k_y)} \times \left(\frac{r^2 - r_0^2 - r_1^2 + r_0^2 r_1^2 / r^2}{r_0^2 + r_1^2} \right) \cos 2\theta \right]. \quad (8)$$

Thus, in this approximation, one has $v_x \sim k_x x/r^2$, and a similar relation for v_y . The location of the rightmost end of the ellipse drawn by the injected dye, $(x_R, 0)$, has then a velocity $dx_R/dt \sim k_x/x_R$ so that $x_R^2(t) \sim k_x t$; similarly, the coordinate of the top end of the ellipse satisfies $y_T^2(t) \sim k_y t$. The ratio of the squares of the major and minor axes of the dyed ellipse, and therefore of the eigenvalues λ_+^2 and λ_-^2 , is equal to the ratio of the corresponding permeabilities for flow parallel to these axes:

$$\left(\frac{\lambda_+}{\lambda_-} \right)^2 = \frac{k_{\perp}}{k_{\parallel}} \quad (9)$$

Another method to obtain this result is to introduce the rescaling $x' = x/\sqrt{k_x}$ and $y' = y/\sqrt{k_y}$, [see de Marsily, 1986] which converts equation (7) to the simpler Laplace equation. The circular boundaries of the flow domain are transformed into ellipses, corresponding to a difficult boundary value problem, but to leading (zeroth) order in the anisotropy they are still circular, and use of the radially symmetric solution of the Laplace equation yields equation (9).

3. Analysis of the Experimental Results

[16] In this section, we apply the procedures described above to determine the permeability anisotropy from the shape of the invasion pattern during the radial injection of a dyed fluid that displaces a transparent one. Several tracer injection experiments were performed for a fixed mean aperture $a_0 = 1$ mm, and different shear displacements in both x and y directions. Reference test experiments with no shear displacement between the fracture walls were also performed to check that aperture variations are statistically isotropic (in this case, a roughly circular shape of the invaded zone is expected for a radial injection). The direction of the maximum flow velocity is associated to the principal axis of the inertia tensor $\mathbf{M}(t)$ with the largest eigenvalue. Figure 4 displays the variation of the angle ϕ , between the principal axis and the x axis, as a function of the radius of gyration of the invasion pattern, R_g . The radius of gyration satisfies $R_g^2 = \text{tr}(\mathbf{M})$ and increases approximately as the square root of time, as required by mass conservation.

[17] The data in Figure 4 correspond to different magnitudes and orientations of the shear displacement $\vec{u}$. In the case of nonzero shear displacements oriented parallel to the y axis (dark symbols), the angle ϕ remains close to zero, with an average over all data equal to 2.5° and variations of the order of $\pm 5^\circ$ around that value. In addition, the curves corresponding to larger amplitudes of the shear displacement ($|\vec{u}| = u_y$, varying from 0.5 to 1 mm) are all very similar. The elongation of the invaded zone is therefore, as expected, globally perpendicular to the lateral shift.

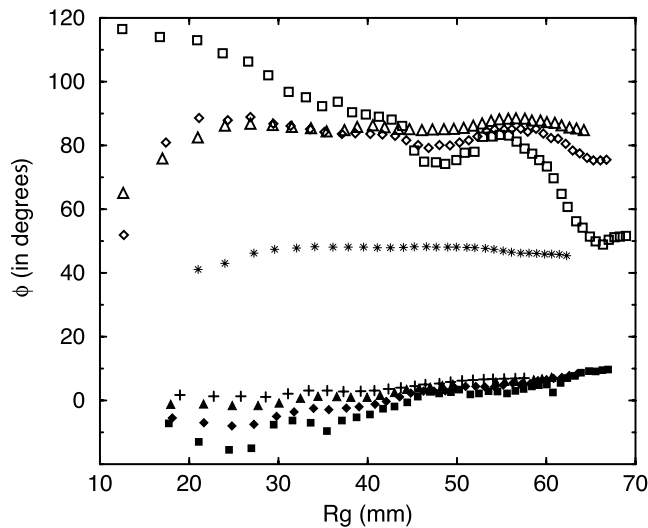


Figure 4. Angle ϕ between the x axis and the direction of maximum flow velocity for different shear displacements. The different symbols correspond to a shear displacement parallel to x and $u = 0.25$ mm (open squares), 0.5 mm (open diamonds), and 0.75 mm (open triangles); a shear displacement parallel to y and $u = 0.25$ mm (solid squares), 0.5 mm (solid diamonds), 0.75 mm (solid triangles), and 1 mm (pluses); and a shear displacement at -45° with x and $u = 1$ mm (asterisks).

[18] Similar results are obtained for displacements parallel to the x axis (open symbols) and at an angle of -45° from it (asterisks). In these cases, ϕ is of the order of 90° and 45° , respectively, indicating again that the invaded zone is elongated perpendicularly to the displacement. This demonstrates that the flow anisotropy is directly connected to the lateral shift, with the preferential flow direction oriented at an angle of 90° with respect to the direction of the shear displacement: these results also show that the observed anisotropy does not arise from any intrinsic feature in the fracture surface itself, but from the shear displacement between the surfaces of the fracture.

[19] These results are in agreement with qualitative observations of the aperture fields, such as that presented in Figure 3: the shear displacement creates ridges in the aperture field, oriented in the perpendicular direction and partly inhibiting the flow. In the absence of such shear displacement we observe, in some cases, a large dispersion of the measured values of ϕ , as expected for a nearly circular invaded zone for which ϕ is undetermined. In other experiments, ϕ takes a better defined value, possibly due to a residual lateral shift.

[20] While ϕ indicates the orientation of the anisotropy, its magnitude is characterized by the ratio of the eigenvalues of the inertia tensor, λ_+/λ_- . The variation of λ_+/λ_- with the radius of gyration R_g of the invaded zone is displayed in Figure 5: it can be seen that λ_+/λ_- reaches an approximately constant value for $R_g > 30$ mm, suggesting that the growth of the invaded zone is self-similar. This asymptotic value of λ_+/λ_- represents then a robust parameter allowing to characterize the anisotropy of the permeability for a given orientation and amplitude of the shear displacement.

[21] Figure 5b shows that for a fixed amplitude of the shear displacement the measured value of λ_+/λ_- , and

therefore the degree of anisotropy of the permeability, depends, in general, on the direction of the shear displacement: e.g., while it is of the order of 1.05 for a displacement $\vec{u}$ along the x axis, it reaches 1.15 when $\vec{u}$ is in the y direction. Note that, on the other hand, when the displacement is applied along a given orientation, the ratio λ_+/λ_- takes similar values in the positive and negative directions. These fluctuations come from the statistical character of the distribution of the ridges that alter the flow, which varies, not only from sample to sample, but also with the orientation of the shear displacement in a single fracture. On the other hand, for a given orientation, the variation of λ_+/λ_- with the magnitude of the shear displacement is smooth, in agreement with the fact that the location of the main ridges remains the same as $|\vec{u}| = u$ increases. These results on the ratio λ_+/λ_- can be converted into permeability ratios by means of equation (9), which allows then for comparisons with the numerical simulations discussed below.

4. Theoretical Discussion of the Relation Between Permeability Anisotropy and Aperture Field

[22] The self-affine nature of the fracture walls affects the aperture field, and has thus an important influence on fluid flow due to the hydrodynamic boundary conditions imposed there. Here, we consider two complementary fracture surfaces with their mean planes parallel and horizontal, which are separated vertically in order to open the fracture and shifted horizontally to introduce a shear displacement. For a vertical separation a_0 and shear displacements u_x and u_y ,

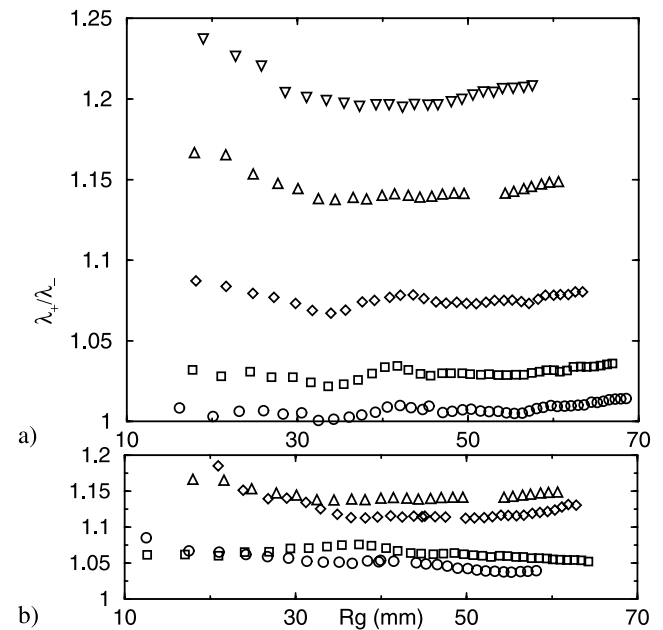


Figure 5. Experimental variation of the ratio λ_+/λ_- as function of the radius of gyration R_g of the invaded zone for a shear displacement $\vec{u}$ oriented along $+y$ and with the following amplitudes: $u = 0$ (circles), 0.25 mm (squares), 0.5 mm (diamonds), 0.75 mm (triangles), and 1 mm (inverted triangles). (b) Variation of λ_+/λ_- with R_g for a constant magnitude of the shear displacement $\vec{u}$ ($u = 0.75$ mm) and oriented in the following directions: $+x$ (circles), $-x$ (squares), $+y$ (triangles), and $-y$ (diamonds).

along the x and y directions, respectively, the local aperture is defined as the vertical distance separating the two walls:

$$a(x, y) = h(x, y) - h(x - u_x, y - u_y) + a_0. \quad (10)$$

[23] The covariance of the aperture, computed using equation (1), exhibits two different scaling regimes [Plouraboué *et al.*, 1995]. Specifically, for length scales smaller than the shear displacement, the aperture field was shown to have self-affine correlations with exponent ζ , whereas, for larger length scales, the covariance decreases, corresponding to a loss of the self-affine character of the aperture field.

[24] The magnitude of the shear displacement thus represents a crossover length scale, and can be regarded as the typical size of features in the aperture field. Using the self-affine scale invariance given in equation (1) we can obtain the scaling of the aperture variance, $\sigma_a^2 = \langle (a(x, y) - a_0)^2 \rangle$, with the magnitude u of the shear displacement, $\sigma_a(u) = Cu^\zeta$, where C is related to the amplitude of the roughness of the fracture wall. Figure 6 illustrates this power law behavior for different orientations of the shear displacement. In all cases, the corresponding power law exponent is $\zeta = 0.75$, in agreement with the Hurst exponent measured in section 2.1 ($\zeta = 0.75 \pm 0.05$).

[25] In the following, we use the ratio of the aperture fluctuations to the mean aperture, $S \equiv \sigma_a(u)/a_0$ as a parameter to quantify the magnitude of the fluctuations in the local aperture that are introduced by the shear displacement. When $S \ll 1$, the aperture irregularities are a small perturbation on a nearly constant aperture field (note that, in the absence of shear displacement between the surfaces, the aperture is constant and equal to a_0 throughout the fracture) and their effect on the flow is negligible; as S approaches 1, the channel becomes quite convoluted and the local aperture variations must be considered in order to correctly describe the flow properties.

[26] A second important result is that the aperture correlation length is anisotropic, being smaller in the direction of the shear displacement than in the orthogonal direction by a numerical factor of $2\zeta - 1 < 1$ [Roux *et al.*, 1998]. The effect of such anisotropy on the flow in rough fractures was studied numerically by Thompson and Brown [1991], who demonstrated that the permeability along the direction of the largest correlation length is enhanced while flow parallel to the direction of the smallest correlation length is inhibited. The effect is related to the occurrence of large ridges oriented along the largest correlation length direction, i.e., in the direction perpendicular to the shear displacement; in fact, such large structures are observed in our system, as can be seen in Figure 3. However, in contrast to the surfaces used by Thompson and Brown [1991], fracture surfaces are isotropic in the present work and the anisotropy of the aperture field results from the shear displacement between the surfaces.

[27] These previous observations motivate us to use a simple model for the permeability anisotropy which assumes that the parallel ridges create parallel flow channels in series with respect to the direction of the shift [Zimmerman *et al.*, 1991]. Flow parallel to the shift is then analogous to an electrical current flowing through resistors in series, and the equivalent hydraulic aperture is $\langle a^{-3} \rangle^{-1}$, whereas

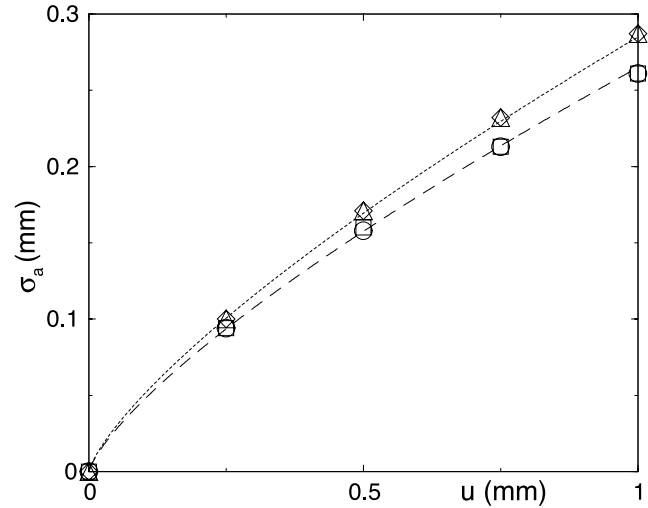


Figure 6. Variance of the aperture field, σ_a , as a function of the magnitude of the shear displacement, u , in directions $+x$ (squares), $-x$ (circles), $+y$ (triangles), and $-y$ (diamonds). Short-dashed line represents $\sigma_a = 0.285 u^{0.75}$, and long-dashed line represents $\sigma_a = 0.265 u^{0.75}$.

flow perpendicular to the shift is analogous to current flowing through resistors in parallel, and the equivalent aperture is then $\langle a^3 \rangle$. Specifically, considering that the distribution of local apertures is Gaussian [Plouraboué *et al.*, 1995], one can estimate the two averages involved, representing upper and lower bounds for the effective hydraulic aperture,

$$\frac{\langle a^3 \rangle}{a_0^3} = 1 + 3S^2, \quad (11)$$

and

$$\frac{\langle a^{-3} \rangle^{-1}}{a_0^3} = 1 - 6S^2 + O(S^4), \quad (12)$$

respectively, in the limit $S \ll 1$. It is interesting to note that equations (11) and (12) accurately describe (up to order S^2) the results derived by [Zimmerman *et al.*, 1991] for both a sine wave and a sawtooth aperture geometry.

[28] Before applying this prediction to the experimental data, we shall first test them on the results of simulations performed on numerical models of self-affine fractures. These numerical simulations allowed us to determine the permeabilities for flow parallel and perpendicular to the displacement vector $\vec{u}$, in contrast to the experimental results that only provide the ratio of these two permeabilities.

5. Numerical Procedure

[29] When the local aperture of the fracture varies slowly along the fracture plane, the lubrication, or Reynolds, approximation can be used to describe the flow in the fracture [Méheust and Schmittbuhl, 2000]. On the other hand, many studies have pointed out that, for natural fractures, this approximation fails to give an accurate description of the hydraulic properties [Yeo *et al.*, 1998;

Nicholl et al., 1999; Dijk and Berkowitz, 1999; Konzuk and Kueper, 2004]. In fact, when the roughness of the walls becomes sizable compared to the mean fracture aperture, e.g. when the surfaces are close to contact, the lubrication approximation fails, and a complete description of the flow field, which accounts for the flow velocity normal to the surface, is needed [Mourzenko et al., 1995; Drazer and Koplik, 2000]. In this work, we use the lattice Boltzmann method to compute the flow field inside the fractures. Previous studies have shown that the lattice Boltzmann technique is particularly suitable to investigate flows in complex geometries [Gutfraind et al., 1995; Gutfraind and Hansen, 1995].

[30] First, a self-affine surface is generated numerically, using a Fourier synthesis method [Drazer and Koplik, 2002], and the fracture pore space is modeled as the region between one such surface and a suitably translated replica. In order to mimic the experimental procedure, the two complementary surfaces are first separated by a fixed distance a_0 in the direction z normal to the mean plane of the fracture. Then, a shear displacement of the upper surface is introduced in the (x, y) plane. The maximum shear displacement investigated in each direction corresponds to the occurrence of the first contact point between the surfaces, and therefore the fracture remains open over its full area. In order to compare the numerical results to the experiments, both the Hurst exponent and the average magnitude of the surface roughness need to be matched to their experimental values: the average fluctuation in surface height over a distance of 1 mm was thus set to 0.25 mm, and the Hurst exponent ζ was set to 0.8. Finally, the mean aperture was set to $a \simeq 1$ mm, which is comparable to that used in the experiments. Using these values, the local height varies considerably over distances of the order of the aperture size, and computing the complete flow field inside the fractures is therefore required.

[31] In order to capture the three-dimensional character of the flow field inside the fractures, large cubic lattices, with as many as $1024 \times 1024 \times 20$ sites, are used in the simulations, which were then run in parallel using a distributed memory computer with over 1000 processors. Hence the lattice spacing δ corresponds to 0.05 mm (the mean aperture $a_0 = 1$ mm corresponds to 20 lattice sites). The Lattice Boltzmann method is then used to compute the flow field in the presence of a pressure gradient and using periodic boundary conditions in the plane of the surface (see Drazer and Koplik [2002] for details of the simulation method). The pressure gradient is applied either along or perpendicular to the shear displacement between surfaces. In addition, and to avoid the possible effects of any intrinsic anisotropy in the generation of the self-affine surfaces, the shear displacements are introduced in both the x and y directions within the mean plane. The Reynolds number in the simulations is $Re = Ua/\nu \sim a^3 \rho \nabla P / 12 \nu^2 \sim 0.4 < 1$ for $a = 1$ mm. Inertia effects are therefore negligible and the flow is governed by the linear Stokes equations, which are invariant under velocity rescaling. Then, by computing the flow rate, we can obtain the permeability of the fracture and its dependence on the magnitude and orientation of the shear displacement.

6. Analysis of the Numerical Simulations

[32] Numerical simulations are performed in configurations reproducing as far as possible those encountered in the

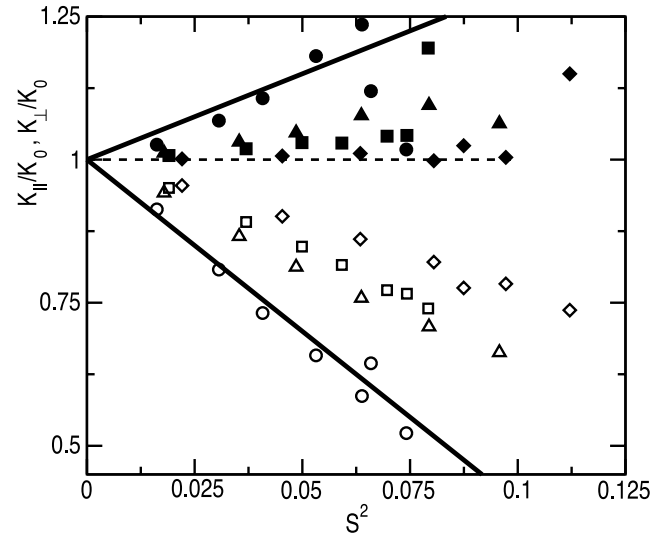


Figure 7. Normalized permeabilities as a function of the normalized aperture variance S^2 from numerical simulations with variable shear displacements ($0 < u < 1.6$ mm) and a constant mean aperture ($a_0 = 1.0$ mm) for two fracture surface realizations. Open (solid) symbols show permeability $k_{\parallel}/k_0$ ($k_{\perp}/k_0$) parallel (perpendicular) to the shear displacement $\vec{u}$. For realization 1, squares show $u_x \neq 0$, $u_y = 0$, and circles show $u_x = 0$, $u_y \neq 0$. For realization 2, diamonds show $u_x \neq 0$, $u_y = 0$, and triangles show $u_x = 0$, $u_y \neq 0$. Top (bottom) solid line is upper (lower) permeability bound from equation (11) (equation (12)).

experiments. Several series of computations were performed on self-affine fracture models for different mean apertures ($0.5 < a_0 < 2.0$ mm) and shear-displacements $\vec{u}$ ($0 < u < 1.6$ mm). The displacement $\vec{u}$ is oriented along directions x or y in the mean plane of the surfaces and the size of the simulated fractures is typically 25.6×25.6 mm. For given values of a_0 and $\vec{u}$, computations are performed on 10 independent realizations in order to estimate the dispersion of the results (only a part of the corresponding data will be displayed for clarity, but all data are consistent with the trends visible on the figures).

[33] The variation of the normalized permeabilities, $k_{\parallel}/k_0$ and $k_{\perp}/k_0$ (where k_0 is the isotropic permeability for $u = 0$, and $k_{\parallel}$ ($k_{\perp}$) is the permeability for flow parallel (normal) to the shear displacement), as a function of the normalized aperture variance S^2 are represented on Figure 7, for a constant mean aperture $a_0 = 1.0$ mm, and for different magnitudes of the shear displacement, which is oriented either along the x or along the y axis. The normalized variance S^2 is used as the horizontal scale since, as already discussed in section 4, it is the relevant parameter to characterize the dependence of the permeability on the shear displacement.

[34] The main result is the fact that both $k_{\perp}/k_0$ and $k_{\parallel}/k_0$ vary linearly with S^2 but in opposite directions ($k_{\perp}/k_0$ increases with S^2 while $k_{\parallel}/k_0$ decreases). Figure 7 also shows that a similar behavior is observed for shear displacements oriented both parallel to the x and y axes: this demonstrates that the permeability anisotropy is induced by the shear displacement and it is not associated to geometrical anisotropies present before the relative displacement of the walls is introduced.

[35] These linear variations of the permeability qualitatively agree with the simple model presented in section 4. In fact, in section 4 we estimated the upper and lower bounds for $k_{\perp}/k_0$ and $k_{\parallel}/k_0$ with S^2 (see equations (11) and (12)). These limiting values are represented by solid lines in Figure 7. As can be seen, the sign and order of magnitude of these variations are the same as those obtained in the numerical simulations.

[36] Another interesting result is the strong correlation between the changes in $k_{\parallel}/k_0$ and those in $k_{\perp}/k_0$: fractures displaying a large reduction of the permeability $k_{\parallel}$ for a given shear displacement generally also display a significant increase of $k_{\perp}$ for the same displacement $\vec{u}$. This is for instance the case for the results displayed in Figure 7, which correspond to two different fractures.

[37] Finally, other simulations were performed in which S^2 is modified by varying the mean aperture from $a_0 = 2.0$ mm to $a_0 = 0.75$ mm, while keeping the shear displacement constant, $u = 1.5$ mm. In this case, the variance S^2 increases when the aperture a_0 decreases. The variations of $k_{\parallel}/k_0$ and $k_{\perp}/k_0$ as a function of S^2 for data sets obtained in this way for three different realizations of the fracture surface are plotted in Figure 8. One observes the same general trend as in Figure 7 for which a_0 is kept constant and u varies: namely, $k_{\parallel}/k_0$ decreases and $k_{\perp}/k_0$ increases or remains constant as a_0 is reduced and S^2 increases. For a given realization of the surface, the variations of $k_{\parallel}/k_0$ and $k_{\perp}/k_0$ with S^2 are linear and both remain within the bounds set by equations (11) and (12). This confirms that the normalized aperture variance, S^2 , is indeed the adequate parameter to characterize the disorder in the aperture field introduced by the shear displacement between the two surfaces.

[38] Numerically, 10 independent self-affine surfaces were used to create aperture fields for different mean apertures and shear displacements. By fitting the average of the permeabilities over all realizations to a quadratic dependence on S we obtain:

$$\left\langle \frac{k_{\perp}}{k_0} \right\rangle = 0.995 + 1.09S^2, \quad (13)$$

and

$$\left\langle \frac{k_{\parallel}}{k_0} \right\rangle = 0.955 - 3.05S^2. \quad (14)$$

Nevertheless, large fluctuations are observed from one realization to another: for some of them, the flow is only weakly perturbed by the wall roughness while, for others, it is strongly distorted and the permeability behaves as predicted by equations (11) and (12).

7. Comparison Between Experiments and Simulations

[39] As discussed in section 2.3, the analysis of the geometry of the invasion pattern in the injection experiments allows one to estimate the ratio $k_{\perp}/k_{\parallel}$ of the permeabilities for flow perpendicular and parallel to the shear displacement $\vec{u}$. The variance S^2 of the aperture can be computed for each experimental setup, i.e., for given values of a_0 and $\vec{u}$, from the profilometer maps of the fracture

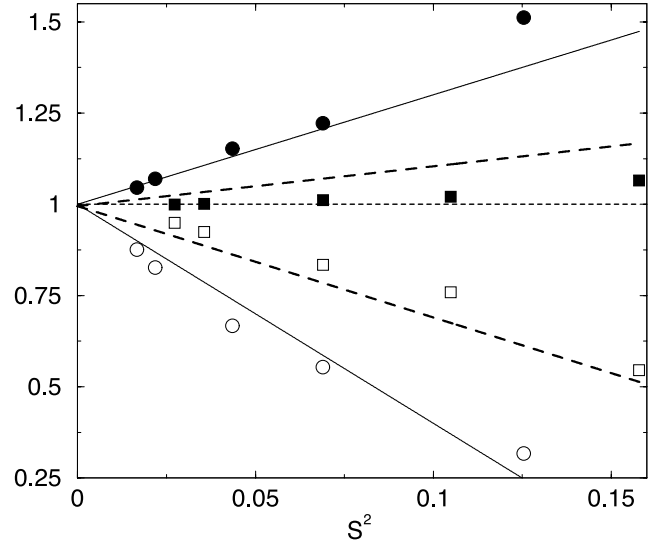


Figure 8. Normalized permeabilities as a function of the normalized aperture variance S^2 from numerical simulations with a fixed shear displacement ($u_x = 1$ mm) and a variable aperture ($0.75 < a_0 < 2.0$ mm) for two fracture surface realizations. Open (closed) symbols show permeability $k_{\parallel}/k_0$ ($k_{\perp}/k_0$) parallel (perpendicular) to the shear displacement $\vec{u}$. Squares show realization 1. Circles show realization 2. Top (bottom) solid line is upper (lower) permeability bound from equation (11) (equation (12)). Dashed lines show linear fits of the variations of $k_{\perp}/k_0$ and $k_{\parallel}/k_0$ with S^2 over all realizations.

surfaces (see section 2.1). It is therefore possible to determine experimentally both $k_{\perp}/k_{\parallel}$ and S^2 for each experiment. Figure 9 displays the variations of the permeability ratio as a function of S^2 , as determined from the shape of the invasion front in a series of injection experiments (see section 2.3). Three sets of experiments are presented, two of them correspond to experiments performed for a constant mean aperture and shear displacements oriented along the x or y direction. No distinction between the positive and negative directions is made since, as pointed out in section 3, the permeability ratio is roughly independent of the shear displacement sign and only varies with its magnitude and orientation. The third set represents experiments performed with a fixed shear displacement, $u_x = 2.0$ mm, and a variable mean aperture a_0 ranging from 1.25 mm to 2.0 mm. As can be seen in Figure 9, the permeability ratios measured with a variable aperture and a shear displacement oriented along the x direction follows the same trend as those obtained with a fixed aperture and a variable shear displacement applied along the x direction.

[40] The permeability ratio increases approximately linearly with S^2 , and remains below the upper bounds provided by combining equations (11) and (12). In addition, the different permeability ratios obtained from the experiments always lie between the strongest and weakest anisotropies observed in the numerical computations. When $S \sim 0.1$, a departure of $k_{\perp}/k_{\parallel}$ from a linear behavior is observed in the numerical simulations. This result is expected since S can no longer be considered to be small compared to 1 and higher-order terms become important. A similar limitation

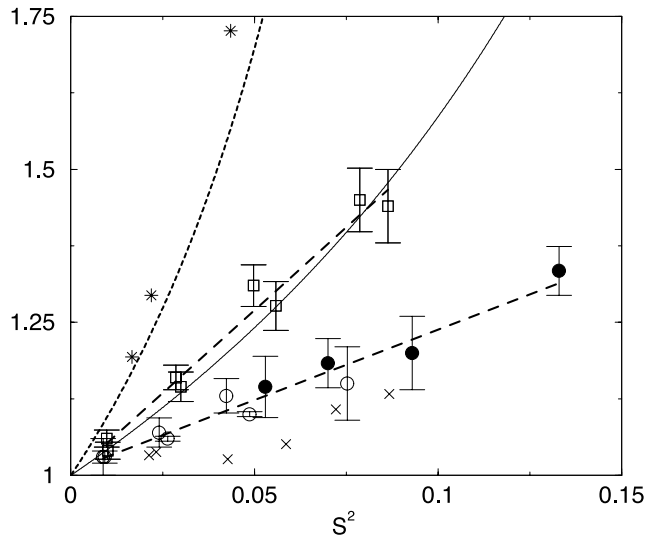


Figure 9. Permeability ratio $k_{\perp}/k_{\parallel}$ as a function of normalized aperture variance S^2 . Squares, open circles, and solid circles show experimental data; stars and crosses show numerical data. Squares show $u_x = 0$, $u_y \neq 0$. Open circles show $u_x \neq 0$, $u_y = 0$, $a_0 = 1.0$ mm. Solid circles show $u_x = 2.0$ mm, $1.25 < a_0 < 2$ mm. Stars and crosses show numerical realizations displaying the strongest and the weakest anisotropy, respectively. Short-dashed line shows upper bound for the permeability ratio estimated from equations (11) and (12). Solid line shows average $\langle k_{\perp}/k_{\parallel} \rangle$ of the permeability ratio over all numerical realizations. Long-dashed lines show linear fits of the experimental data.

applies to the experimental results, in that the method used for the anisotropy characterization is strictly valid for weak anisotropic permeabilities (see discussion in section 2.3).

[41] Finally, we also present in Figure 9 a linear fit to the experimental data. The strongest effect is observed for a shear displacement in the y direction and a linear regression indicates a linear increase of $k_{\perp}/k_{\parallel}$ with S^2 with a slope of 5.4 ± 0.3 . For a shear displacement in the x direction, the permeability ratio also increases as S^2 but with a slope of 2.3 ± 0.15 .

8. Conclusions

[42] The present study shows, both experimentally and numerically, that the permeability of a fracture made of two complementary surfaces with a shear displacement $\vec{u}$ is lower for flow parallel to $\vec{u}$ than perpendicular to it. More precisely, the parallel permeability ($k_{\parallel}$) decreases significantly while the perpendicular one ($k_{\perp}$) increases or remains constant as the magnitude of the shear displacement increases.

[43] Qualitatively, this result appears to be related to the appearance of preferential channels of larger local aperture oriented perpendicular to $\vec{u}$: these channels are indeed visible on aperture images reconstructed from experimental (or numerical) profilometry maps. Assuming such a geometrical structure allowed us to develop a simple theoretical model to predict the relation between the permeability variations and the normalized variance S^2 of the aperture field. The permeability for flow parallel (perpendicular) to $\vec{u}$ was then shown to decrease (increase) approximately line-

arly with S^2 . The absolute magnitude of the variation is predicted to be two times larger for the parallel flow. We showed that these predictions are qualitatively correct and provide a good estimate for the permeability variations and for the ratio of the permeabilities perpendicular and parallel to the shear displacement. In particular, both experiments and numerical simulations exhibit a linear variation of the permeability ratio with S^2 , showing that in fact, the variance S^2 is a key parameter of the process. However there is a significant dispersion in the absolute value of the constants in these linear relations from one sample to another.

[44] Further studies will be needed to understand the origin of the dispersion of the numerical values of the permeability variations and uncover the other characteristic geometrical parameters which may be involved in the process (finite size effects may be a limitation, particularly for the present simulations). Another important issue for practical applications is whether an estimation of the aperture variance can be obtained, for a given shift $\vec{u}$ and mean aperture of the fracture a_0 , without needing to map the full surface of the sample. Mean aperture values used in our experiments and simulations are comparable to those corresponding to near-surface fractures, but substantially larger than for deeper fractures. We have shown, however, that the scaling behavior found for permeability variations only depends on the fluctuations of the local aperture normalized by its mean value, and not on the absolute value of the aperture. We therefore expect the same behavior for open fractures located far from the surface. Of course, since the stress on the fractures increases with depth, the contact area between opposite surfaces can increase substantially with depth, and could modify the analysis.

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